

Element I

Build and Test Evaluation

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Problem Statement

Medical institutions lack a cost-effective, risk-free method for surgical training and other medical processes. Certified cadavers are the best training material for students, but are in great shortage worldwide, and pose ethical and practical issues. Due to said scarcity, organizations such as government agencies allocate a great proportion of their budget towards acquiring these materials: Medicare spends ~\$180,000 per medical resident each year through subsidies to universities, totaling over \$25 billion.¹ Overall, inefficient distribution and lack of product availability hinder surgical trainees worldwide, thus a more accessible solution is required.

I. Prototype Building

The first step taken in the assembly of the prototype was to 3D print and then assemble the servo assemblies. This required cutting each spiral spring to an appropriate length and folding the edge of the spring to wedge into the knob of the potentiometer. The Potentiometer tensioners were then put on the potentiometers and the springs were then inserted.



Figure 1.1: Servo tensioner with spring wedged into potentiometer.

The String was then wound around the string spool, which was then placed into the slit on the potentiometer, essentially connecting both pieces. After, the string cover was placed around the string spool to prevent any tangling, and then the whole assembly was placed into the individual servo mount.



Figure 1.2: Completed servo mount.

Incremental Testing #1

The first incremental test was performed to confirm that the spring, string, servo, potentiometer, and 3D prints all mesh correctly and properly. This can be tested by seeing if the potentiometer is able to rotate approximately 330 degrees when the string is pulled on. This test is a

¹ "The Costs of Training Internal Medicine Residents in the United States." The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

success if the potentiometer achieves a full range of motion from the minimum to maximum value when the string is pulled, and if the potentiometer returns to its base position when released, all being conducted with minimal force.

The first result of this test was an immediate failure, as the entire part required an immense amount of force for the potentiometer to rotate, and the potentiometer only rotated about a half rotation. This was caused by many of the 3D printed parts rubbing against each other causing unnecessary friction. Additionally, the springs needed to be inserted in a particular way, where the potentiometer was at its left or rightmost position for the springs to rotate the potentiometer all the way back.

In order to remedy these issues each of the 3D printed parts were revised and the test was repeated multiple times. After about a week of modifying the parts, nearly all the friction within the assembly was fixed, and a final version was decided upon. Five copies of each part were then printed such that they could be assembled into five servo assemblies.

The five copies of the servo assemblies were made to include a set screw in the servo mount to act as a hard stop. The servos mounts were then put into place on the string guide, and were screwed in with M3 nuts and bolts.

The servo mounts were attached onto the string guide to also test the clearance for the screw holes and strings to flow correctly.



Figure 1.3: 3D printed servo assembly parts with red PLA.



Figure 1.4: Individual servo mounts with string testing.



Figure 1.5: 3D printed string guide.



Figure 1.6: Bottom of string guide with string threaded through.

Another part of the assembly was the battery mount, which was first modeled in 3D. To be confident in the accuracy and precision of the model, a negative mold (concave-shaped mold that is “cut” into the outer shape of the object) was printed of the battery and the gap between them was determined to ensure easy removal but still a secure fit. The actual mount of the battery was then redesigned and printed.

Then, the top and bottom half of the battery assembly were joined together with a mixture of hot glue and super glue. The batteries were then inserted into the holder with USB-C Female ports attached to them. These were glued in place and one wire was soldered between the two 5 volt terminals while another was soldered between the two ground terminals.



Figure 1.7: Battery with clearance test piece.

Incremental Testing #2

The purpose of the second incremental testing is to make the tolerance between the battery mount and batteries secure enough for it to not fall out, but not too tight that it can't be easily taken out. The test is considered a success if the battery is able to be plugged in and unplugged with only two fingers, and if shaken, does not fall out.

This test was initially a failure as the battery was unable to be pulled out without additional tools or several tries. This was remedied by using a dremel and sandpaper to hollow out the places of contact, which reduced the amount of friction between the battery and the holder. After repeating the process and checking the alignment a few times, the test was considered a success.



Figure 1.8: Test if battery fits properly into battery mount.

The battery mount was then connected to the rest of the glove with M3 screws and was super glued in place to increase the rigidity.

Once the mechanical aspects of the glove had been assembled, and the electrical systems needed to be introduced.

This started with soldering wires between each of the negative and positive terminals on all of the potentiometers, and then the

data wires were connected for each. This started with soldering wires between each of the negative and positive terminals on all of the potentiometers, and then the data wires were connected for each.

Once finished, a black wire was connected from the negative wires of the potentiometers to the ground on the USB-C female ports. The green wire is connected from the positive wire of the potentiometer to the 3.3V output on the Arduino Nano. Each data wire was then connected to the corresponding pin.



Figure 1.9: Picture of Partially Wired Electronics

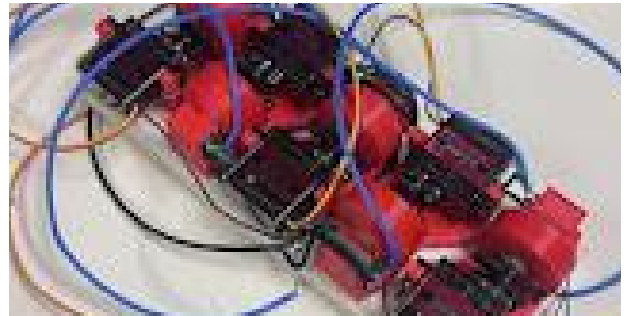


Figure 1.10: Picture of Partially Wired Electronics w/
Data Wire

Incremental Testing #3

The purpose of the third incremental test was to ensure that the electrical system for the potentiometers worked correctly. This test is considered a success if each potentiometer reports a corresponding voltage for how far its string is extended.

This test was initially a failure, as each potentiometer was outputting a value of 4095 regardless of how far the string was pulled. This issue was resolved by rewiring incorrect negative to the arduino. When fixed, four of the potentiometers reported correct values, but the last potentiometer was displaying 0. The issue we found was that part of the solder had come off and was creating a short between the negative and data terminal of the potentiometer. By resoldering these two pins, the last potentiometer was fixed and the test was considered a success.

Now that the electrical system for the potentiometers was done, the electrical system for the servos needed to be done.

To accomplish this, each servo had the end of it cut off, and then each wire was separated. Each of the black wires were soldered together and then connected to one of the ground terminals on a potentiometer. Then, each of the red wires were soldered together, and then with one more red wire they were connected to the 5V terminal on the USB-C female breakout boards.

Then each yellow wire (data) from the servos was connected to a different colored wire with a female dupont connector on it. Each data wire was then plugged into its corresponding spot on the Arduino Nano.

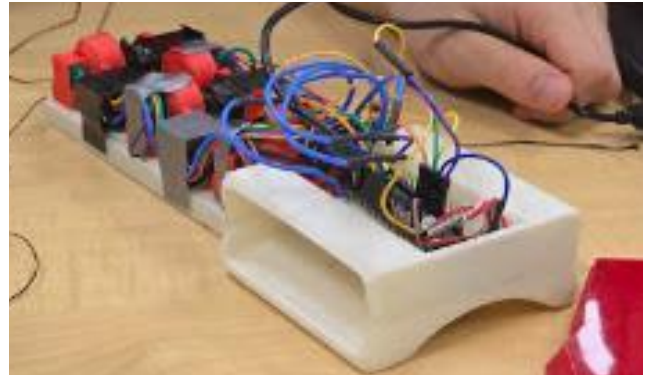


Figure 1.11: Fixed and merged electrical systems.

Incremental Testing #4

The purpose of the fourth test was to determine whether the electrical system of the servos was wired correctly. This test included sending 3 different signals to each servo, each one corresponding to a position of 0, 90, and 180 degrees. This test is considered successful if each servo is able to repeatedly spin to the correct position when the signal is sent.

As predicted, the first initial test was unsuccessful, as when the signal for 90 degrees was sent, no servos moved. When the signal for 0 was sent, each servo rotated in the same direction continuously, and rotated in the exact opposite direction when the signal for 180 was sent. The problem was horribly obvious(*mmm, so devious!*), as the servos that were being used were continuous rotation servos, not positional servos.

The quickest way to resolve this issue would have been to simply cut the wires right next to the servos and solder the new servos directly to those cut ends. However, it was decided that the best course of action was to rewire the entire electrical system with better technique. This included using custom dupont connectors crimped onto the wires being used throughout the glove.

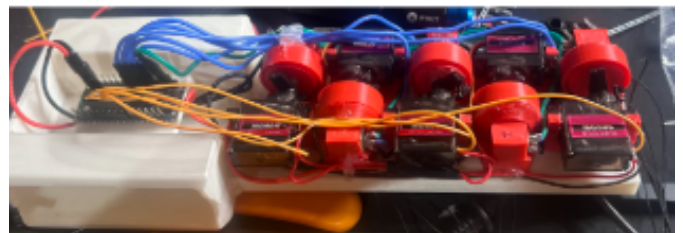


Figure 1.12: Resoldered servos on glove.

Incremental Testing #5

The purpose of this test was the same as tests #3 and #4, but this time both of them combined, validating the entire electrical system combined.

While all of the servos responded correctly to the signals, only four of the potentiometers were working correctly. It was determined that the potentiometer that wasn't working was broken, and after desoldering and replacing it, the entire electrical system ultimately functioned correctly. Additionally, to prevent any tangling, the wires were reorganized to run more along the sides of the glove rather than largely in between the servo mounts, which helped keep the main rotation points clean and spacious.

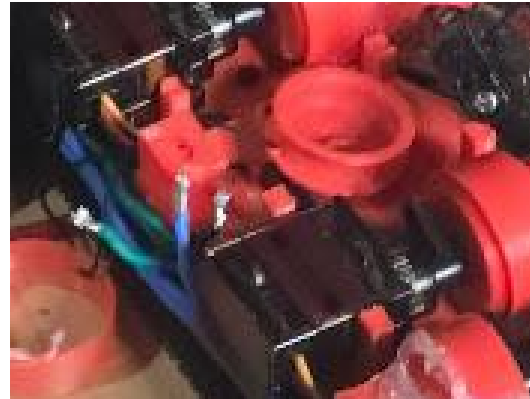


Figure 1.13: Desoldered potentiometer.

Incremental Testing #6

The purpose of this test was to determine if the entire electrical system is able to work in conjunction with each other. This is done with arduino code which relates the position of the servos to the rotation of the potentiometers. Thus when you pull on each string, the corresponding servo will rotate.

This test was initially a failure, as it was found that the servo arms which are attached to each servo are long enough that they interfere with each other. The problem was easily fixed by simply cutting a small amount of the end of each servo arm.



Figure 1.14: Cut off servo arms.

Incremental Testing #7

The purpose of this test was to ensure compatibility between the glove, Arduino IDE, and Unity. The way this was tested was by writing code which relates the amount each string is pulled out to how much each finger of the digital glove model is contracted.

This test was not an initial success as when a string was pulled it took around a minute before the digital model updated. This was fixed by adjusting the arduino so that it outputted less data points per second, so that Unity wouldn't get backed up. The baud rate (a measure of how many times the Arduino updates itself per second) within the Arduino IDE was also updated to reduce latency. After many iterations of the code, the digital model responded accurately with only $\sim 7 \pm 6$ ms delay, and the test was considered a success.

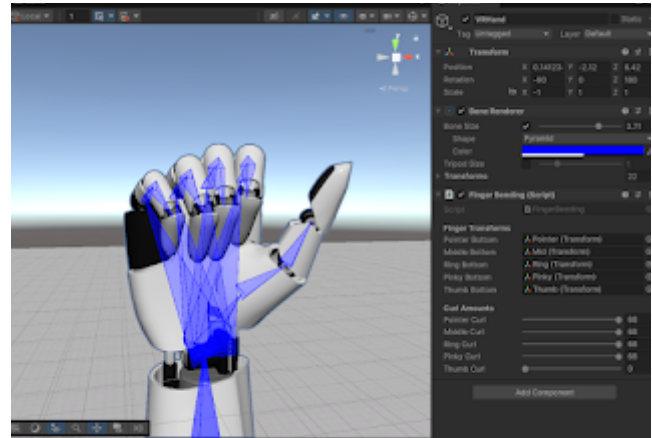


Figure 1.15: Digital model of right hand curled up.

The next step was to allow the glove to mount and affix to the hand itself. This began with printing out a dozen or so flexible caps which fit on the top and bottom of each finger. Both pieces are then attached together with flexible material to keep the self-adjusting quality. They are attached by a mixture of plastic welding and super glue.

Incremental Testing #8

The purpose of this test was to determine if the size of each finger cap was acceptable. This was done by putting on the underlying glove, putting on each finger cap, and wearing it for 5 minutes. If the cap did not come off or cause the wearer any pain or discomfort, then that cap is considered successful.

While the pointer, middle, and ring finger all succeeded in this test, the thumb and pinky finger both failed. The pinky finger was far too small to be put on and the thumb was the exact opposite.

The solution to this problem was to cut off the elastic connectors for the two caps and adjust the length of them so that the cap would fit better. After this, the test was run again and the test was considered successful.



Figure 1.16: Testing finger caps to see if they are well-fitting.

After the caps were attached, a string guide was attached to every joint on each finger to prevent string from rubbing against the fingers and knuckles, and also to keep the string on top of the finger and not slide off. To ensure that every string guide was properly mounted, an attempt to pull it off was given, and if it didn't disconnect, it was considered acceptable.

The next step was to make a way for the rest of the glove to mount to the arm. This included attaching velcro strips along the length of the glove on both sides.



Figure 1.17: Testing finger guides at marked locations on the hand.

Incremental Testing #9

The purpose of this test was to determine if the velcro is enough to hold the glove onto the arm. This was done by wrapping the arm with velcro which is attached to the electrical mount. The tester then moves their arm around and sees if the glove is too loose. As long as the glove doesn't move around on the arm and doesn't cause the user any discomfort, this test would be considered successful.



Figure 1.18: Velcro straps attached to glove.

This test was not initially considered a success as the glove did move around on the arm, however after adding two more strips of velcro around the arm and tightening the strips a little bit, the test was considered a success.

Subsequently, it was decided to perform a preliminary test fitting to make sure that everything fit correctly and to see how well things were working. In order to do this, each string needed to be threaded through its designated spots with the user's hand in the glove. The strings were then pulled taught and glued to the corresponding finger cap.



Figure 1.19: Fully assembled glove worn by Ryan.

Incremental Testing #10

The purpose of this test was to determine if the user has enough range of motion while using the glove. This test is considered successful if the user is able to go the entire range of motion in the fingers, wrist and elbow.

The first pass of this test was not successful, as the wrist was not able to rotate more than 15 degrees above parallel. This was caused by the hand mount colliding with the servo mount. To fix this, about an inch was cut off the base of the wrist mount so that it wouldn't collide with the rest of the glove. This fix freed up most of the range of motion within the wrist. However, if a user does not have a big enough forearm, the battery mount will prevent some range of motion at the elbow. This problem has not been fixed, and there are no plans to fix it for this prototype as the size of the fix is outside the scope of this prototype.

Incremental Testing #11

The purpose of this test was to determine if closing the users hand is able to adjust the values outputted by the potentiometer. This test is considered successful if the amount that each finger contracts directly correlates with a different output by the servos, and if that output is consistent throughout.

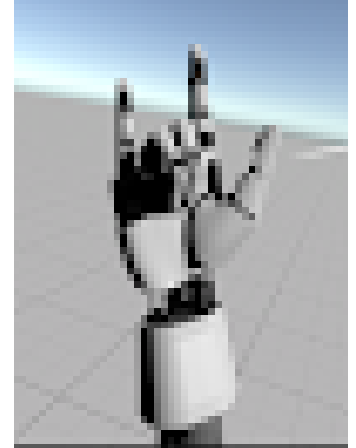
Initially this test was a failure, as only the pointer, middle, and ring finger were able to adjust the values of the potentiometers. However, as the test continued both the middle and ring finger stopped working. They stopped working because the connection between the string and the finger caps had loosened enough that the string was no longer rigidly attached to the caps. On the other hand(*pun intended*), the pinky and middle finger weren't working because it required too much force for them to be pulled fully.

A silicone lubricant was used to lubricate each of the servo assemblies, which significantly reduced the friction and fixed the problems occurring with both the thumb and pinky finger. However, the middle and ring finger problems revealed a need for a better mounting solution. This came in the form of an additional string guide on each of the finger caps. This avoided the difficulties that arose when trying to glue things to the flexible material.



Figure 1.20: String threaded through each guide and connected to finger caps.

Finally, velcro was added to the controller holder to make the mount more rigid. In addition, the actual quest controller holder was added onto it with a mixture of hot glue and superglue. Hot glue ridges were added around the rim and inside to create enough friction so the controller would not slide out.



II. Prototype Software Building

This section delineates the progress toward the digital aspect of the glove.

Firstly, a digital hand must be made to then connect with the motions and hardware of the real prototype. Originally, a model was to be made by the team to resemble the finished product. However, that yielded mixed results, inoperable in the digital environment and tedious given the timeframe. With that in mind, the team pivoted to using a pre-existing hand model we had previously vetted and liked. This strategic decision saved time and allowed me to focus on refining functionality.

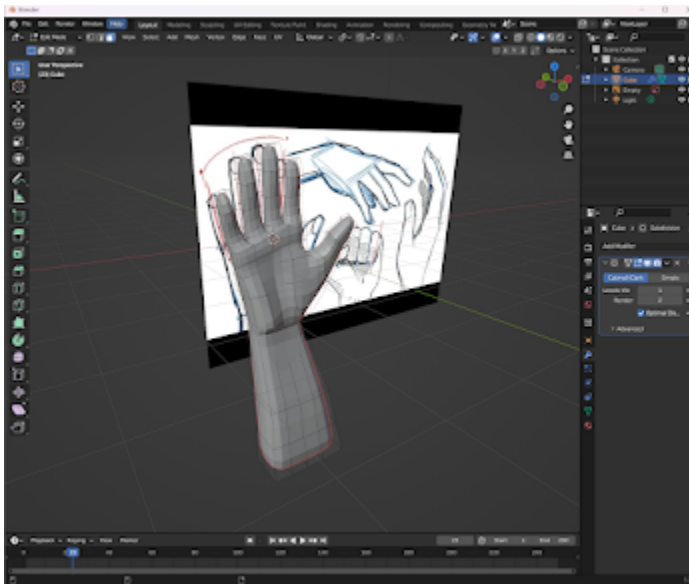


Figure 2.1: A preliminary original model that was scrapped



Figure 2.2: The pre-existing hand model, with textures. The black lines are the 'bones' used to animate the hand.

Imported into the app Blender, a rig was created to animate the hand. Initially the team applied an armature deform, which (aptly named) treated the mesh as malleable. This method bent the entire

object in a way that didn't suit the discrete parts of the hand. The team explored weight painting to control how different parts of the mesh respond to bone movement. This technique uses gradient weights to dictate how much influence a bone has over the mesh—useful, but more detailed than needed at the moment. Eventually, a more direct method was set: parenting individual objects to their corresponding bones. This allowed me to control finger segments more intuitively and cleanly, with automatic movement following the bone structure. These 'bones' were connected to each fingers' meshes (shape) to move using forward kinematics, a staple in Blender's animation system. Textures were 'mapped' or added to the model and sent into Unity, the basis for our VR simulation game.

Unity didn't recognize the bones automatically, so they were manually attached using a "bone renderer" script. This required some custom stitching to connect everything properly. Then, the FingerBending.cs script was made: this script would access the finger joints and curl the fingers from a value of 0 (unflexed) to 68 (completely close). To prototype with one finger (the pointer), it was first set up as a test rig. Then, a variable called curlAmount was defined, which rotates each segment by a consistent offset from the original position. Then, a method was made to control finger curl dynamically—this sets up compatibility with potentiometer input later on. This value would then be applied to each joint with a scale factor based on the type of joint.



Figure 2.3: A working animated hand model, with four main fingers curled to the value 68.

The model was uploaded as the controller into an XR Origin Template scene (a basic template with already implemented VR code), which had its issues with the Oculus Quest we were using and the outdated template within Unity's conflicting update systems, but the right controller was attached to the model, tracking both position and rotation data and keeping the template clean. Then the position of the hand was tweaked to fit how the controller would actually be oriented on the glove (sideways).

Here, a reflection probe and lighting environments were made, along with a table and objects to test collisions later on and to make the digital world more appealing.

Once the servos were completely fixed, Unity to Arduino communication was established through the serial IO ports, speaking to each other on one lane essentially. The Arduino would output the potentiometer readings, mapped to 0-68 based on the finger curling. From there, Unity collects this data via the Serial.IO package, and assigns each value it reads to every finger. Originally, some issues arised with authority and access issues with this kind of communication. Because of the ESP board, not exactly native in the Arduino family, Thus, additional access permissions were enabled by changing the MONO based backend and switching the API Compatibility level to .net framework instead of 2.1, different packages to communicate with the Arduino better, and workarounds were made the stabilize the communication. This included enabling DTR and RTS, control signals to open the communication gates. This was the most difficult part in the programming process.

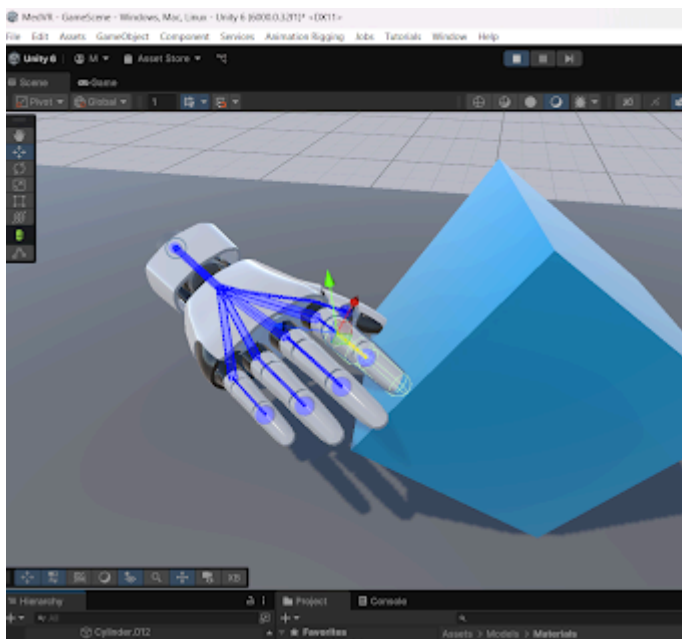


Figure 2.4: The hand model colliding with a TestCube object.

From there, Capsule Colliders, a cylindrical shaped object, were attached, wrapping around each finger part. These were then attached to Rigidbody components. The Rigidbody, essentially, makes the object rone to physical interactions with other RigidBodies, as Unity knows to include those in collision calculations. Capsule Colliders determine the shape of the object that is used for collision data and processing. Using the mesh itself for such would be extremely intensive, and by simplifying the geometry to invisible colliders, the program runs more efficiently. The glove programming then took a break to wait for the actual hardware to be completely assembled as one piece. In this time, a backup was created of all the code on GitHub to be collaboration friendly.

Finally, the servos were coded to give force feedback. Essentially, HandCollision and HandCollisionManager scripts were made to handle communication between Unity and Arduino. HandCollision was added as a component to each Capsule Collider, which would report to the Manager when a collision was detected. This manager would keep each finger's collision data as 0 or 1: 0 is no touch, 1 is touching. Whenever, a significant change was made in this state (that is, no glitchy collisions were made and it was a purposeful touch) the Manager would then send data back as a string to the Arduino. The Arduino would then lock the servo at the exact angle the potentiometer of that finger was at right when the collision occurred. Otherwise, it releases to give full range of motion. This would limit further extension, simulating actual touch. This system was definitely not optimized, and thus keys were used to differentiate data. The connection between

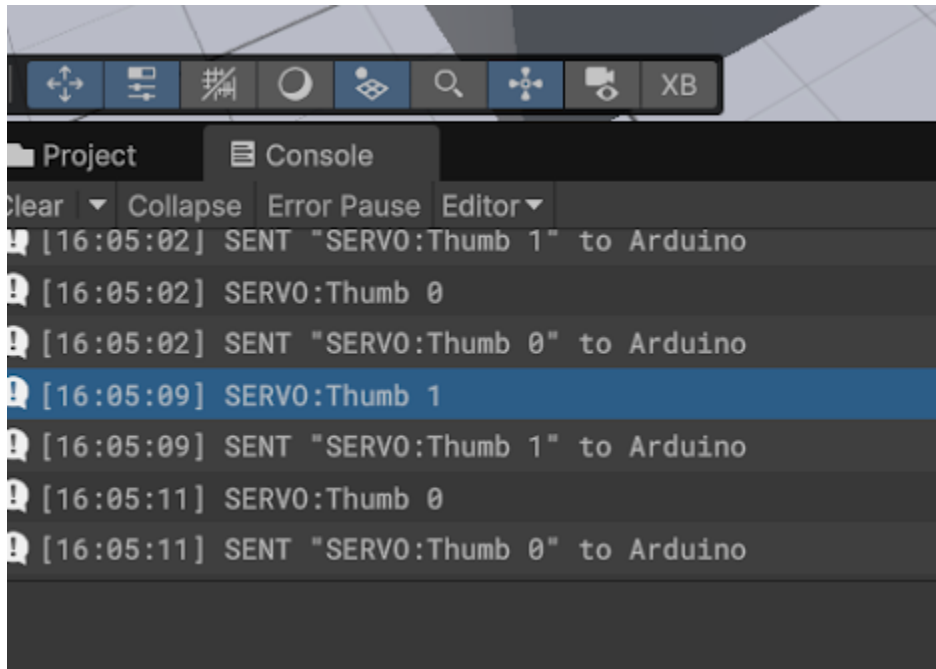


Figure 2.5: A preview of debug log messages indicated that SERVO messages were being sent to the Arduino.

hardware and software was through one lane where both Unity and Arduino send data to the same monitor. So, each message was prefaced with either “POT” for potentiometer data or “SERVO” for servo data. This would allow for clearer communication, eradicating lag. Finally, when an object did in fact get gripped by the hand (that is, the thumb and multiple fingers are touching an object and are curled) it would get anchored to the hand, simulating an actual grab, completing the servo mechanism loop. This concludes the full package of software.

III. Testing and Results

After the haptic VR glove was fully assembled, three tests were conducted to test the effectiveness and feasibility of the glove: restrictive force on fingers, position and rotational accuracy, and weight/center of mass.

Test 1: Restrictive Force on Fingers

Purpose: To determine the force exerted by the servo depending on varying input signals.

Test Procedure:

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Have the participants put on the glove, and put the servos at 90 degrees
5. Record each finger is able to over power the servo.
6. Put a scale on a counter and have the participant rest their arm on the same table with their hand resting on the scale.
7. While preventing the hand from moving upwards, have the participant apply as much force as they can into the scale through just one of their fingers.
8. Record this number in a data sheet and repeat with each finger.
9. Convert the number of grams to kilograms by dividing by 1000 g/kg.
10. Convert the number of kilograms to newtons by multiplying by 9.81N/kg
11. Repeat this experiment with different people, as well as with each finger.



Figure 3.1.1: Image showing how weight was measured using a scale.

Results:

Ability to contract finger despite resistance from servo

Finger	Tester 1	Tester 2	Tester 3
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Thumb	No	No	No
Pointer	No	No	No
Middle	No	No	No
Ring	No	No	No
Pinky	No	No	No

Figure 3.1.2: Table regarding ability to contract fingers

Figure 3.1.3: Weight reported by Scale in grams

Finger	Tester 1	Tester 2	Tester 3
Thumb	3900 g	4300 g	4100 g
Pointer	3100 g	3500 g	3400 g
Middle	3000 g	3200 g	3200 g
Ring	2100 g	2000 g	2300 g
Pinky	1400 g	1500 g	1800 g

*Scale reports weight to nearest 100 grams resulting in error of ± 50 grams

$$\text{Force} = \text{Weight in Grams} / 1000 * 9.81$$

Figure 3.1.4: Weight reported by Scale in grams converted into Newtons

Finger	Tester 1	Tester 2	Tester 3
Thumb	38.3 N	42.2 N	40.2 N
Pointer	30.4 N	34.3 N	33.4 N
Middle	29.4 N	31.4 N	31.4 N
Ring	20.6 N	19.6 N	22.6 N
Pinky	13.7 N	14.7 N	17.7 N

*Data rounded to 0.1 Newtons, with an error of ± 0.49 Newtons

Data Analysis:

No one who tested out the glove was able to over power the servos, which means that the force applied by the servos is greater than the forces recorded by the test above. However it can give a lower bound for the amount of force. Seeing as the force should be the same between each

finger as the mechanical system are identical between them. Thus the highest value above will be used to calculate the lower bound for the amount of force applied by the glove. The highest value above is 42.2 Newtons, so this is the lower bound for the amount of force the glove is able to apply.

Test 2: Positional Accuracy

Purpose: To measure the variance in glove movement between the real and digital world.

Test Procedure:

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Begin motion tracking using the Test Application.
5. Have the participant place their hand at a set zero position in the real world.
6. Mark down the digital position of the zero.
7. Have the participant move the controller to a set position
8. Record the virtual position in the data sheet
9. Repeat for multiple different positions.
10. Measure the deviation in position between real and digital environments.
11. Repeat this for three trials.



Figure 3.2.1: Image showing the testing setup for Test 2

Figure 3.2.2: Trial 1 Raw Data

Real World Position (in)	Trial 1 X (m)	Trial 1 Y (m)	Trial 1 Z (m)
0	-1.0241537	0.69195888	0.13183932
6	-0.8717556	0.69255	0.1284159
12	-0.7191283	0.6925911	0.1269658
18	-0.5652421	0.693876	0.1240121
24	-0.4068552	0.6965472	0.1215306
30	-0.2544618	0.6995645	0.1223815
36	-0.1029074	0.7015074	0.1172988

Figure 3.2.3: Trial 2 Raw Data

Real World Position (in)	Trial 2 X (m)	Trial 2 Y (m)	Trial 2 Z (m)
0	-1.21330196	0.72186196	0.019953153
6	-1.0614638	0.7221941	0.0166358
12	-0.9105031	0.7233226	0.0136463
18	-0.7555941	0.7230346	0.0083935
24	-0.6000762	0.7253962	0.0090082
30	-0.4500473	0.7278836	0.0036048
36	-0.2925456	0.7282767	0.0020635

Figure 3.2.4: Trial 3 Raw Data

Real World Position (in)	Trial 3 X (m)	Trial 3 Y (m)	Trial 3 Z (m)
0	-1.25772614	0.7188783	0.5524878
6	-1.112758	0.7208439	0.5028106
12	-0.968443	0.7205367	0.4551796
18	-0.824045	0.7226887	0.4062093
24	-0.677658	0.7256215	0.3531812
30	-0.5291318	0.7273654	0.3060853
36	-0.3879173	0.7306719	0.2544246

$$\text{Distance} = \text{Sqrt}((x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2)$$

X_0 = Value of X at 0 in

Y_0 = Value of Y at 0 in

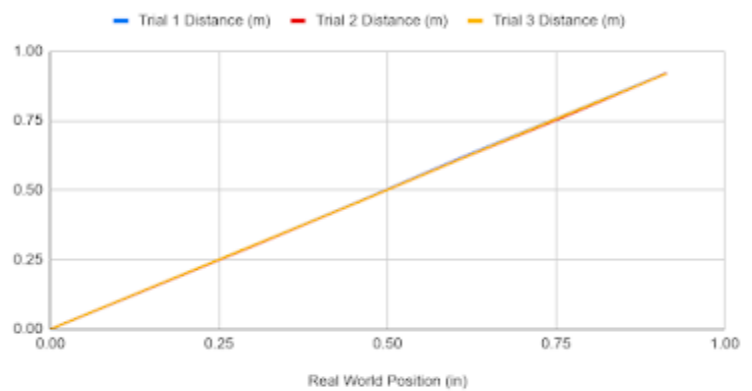
Z_0 = Value of Z at 0 in

Figure 3.2.5: Combined Distance Data for Each Trial

Real World Position (in) (m)		Trial 1 Distance (m)	Trial 2 Distance (m)	Trial 3 Distance (m)
0 in	0 m	0	0	0
6 in	0.1524 m	0.1524376926	0.1518747575	0.1532561562
12 in	0.3048 m	0.3050649858	0.3028680562	0.3052152866
18 in	0.4572 m	0.4589823497	0.457855311	0.457702141
24 in	0.6096 m	0.6174016201	0.6133336087	0.6133902824
30 in	0.7620 m	0.769787579	0.7634534726	0.7691787407
36 in	0.9144 m	0.9214105199	0.920952476	0.9195369369

Trial 1	$r = 1$ $R^2 = 1$
Virtual = 1.01033 * Real - 0.00119518 (m)	
Trial 2	$r = 1$ $R^2 = 1$
Virtual = 1.00686 * Real - 0.00171752 (m)	
Trial 3	$r = 1$ $R^2 = 1$
Virtual = 1.00737 * Real - 0.00081338 (m)	

Figure 2.2.6: Chart comparing Real Position to Virtual Position



$$\text{Percentage Error} = \frac{|\text{Measured} - \text{Theoretical}|}{\text{Theoretical}}$$

Figure 3.2.7: Combined Distance Data for Each Trial

Real World Position (in) (m)		Trial 1 Error %	Trial 2 Error %	Trial 3 Error %	Mean Per Position
6 in	0.1524 m	0.02473264649	0.3446472786	0.5617822515	0.3103873922
12 in	0.3048 m	0.08693758946	0.6338398213	0.1362488801	0.2856754303
18 in	0.4572 m	0.3898402669	0.1433313545	0.1098296213	0.2143337476
24 in	0.6096 m	1.279793322	0.6124686208	0.6217654923	0.838009145
30 in	0.7620 m	1.021991991	0.190744434	0.9420919561	0.718276127
36 in	0.9144q m	0.7666797813	0.7165869561	0.5617822515	0.6816829963
Mean per Trial :		0.5949959329	0.4402697442	0.4889167421	0.5080608064 %

Figure 3.2.8: Chart Showing Percentage Error By Test #

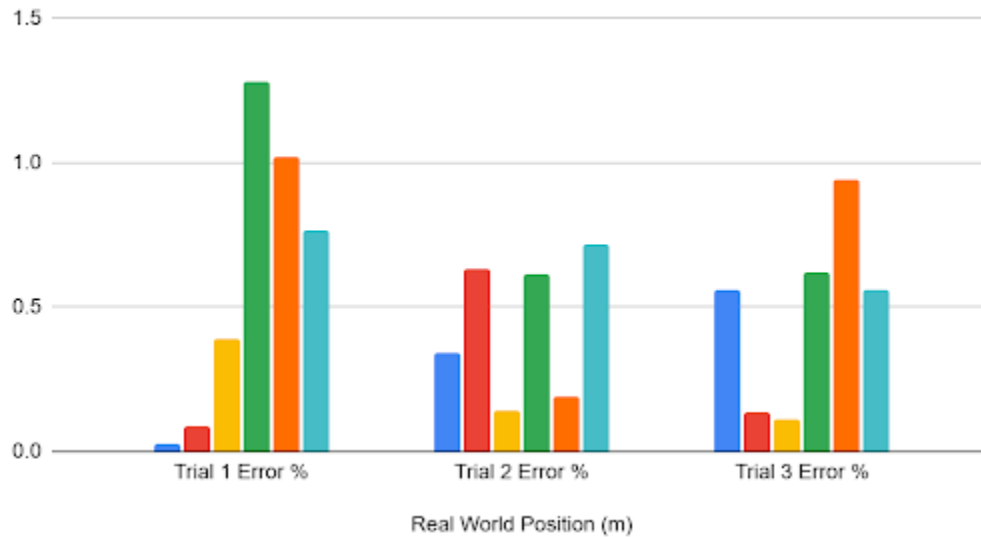
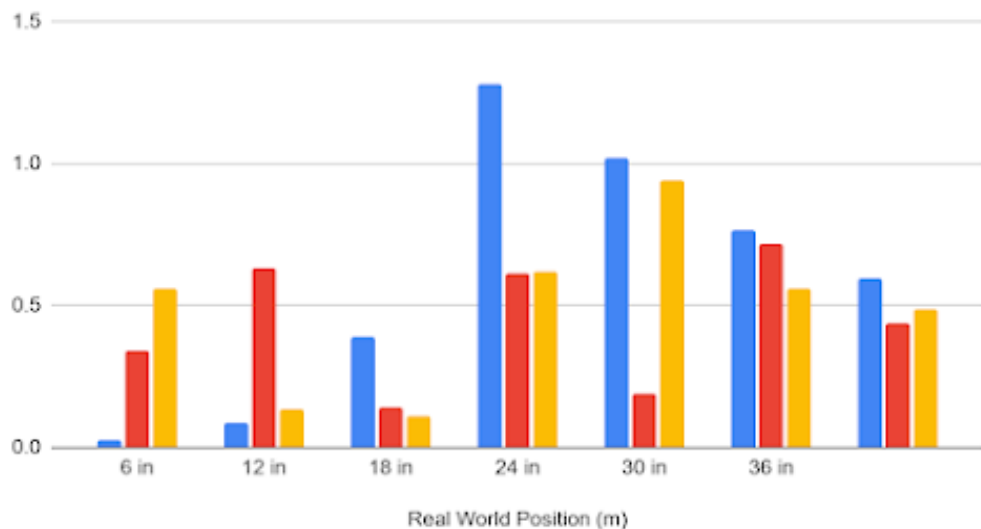


Figure 3.2.9: Chart Showing Percentage Error By Position



Data Analysis:

The average percentage error of over these three tests is approximately 0.5%, which is phenomenal for the use case. This percentage error is small enough that it could come from human error. It also appears that the data which came from Tests 2 and 3 appear to be more accurate than test 1, as test 1 has a major error which occurred at 24 and 30 inches. Looking at the data when grouped by position, the error remains roughly the same, which signifies that the error isn't amplified by increasing the position.

Test 3: Weight and Center of Gravity

Purpose: To determine the glove's center of mass and weight to ensure comfortable usability in a VR environment.

Test Procedure:

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Setup the balancing rig
5. Record the weight on the scale in a data table
6. Adjust the distance of the scale to the pivot
7. Repeat for multiple locations



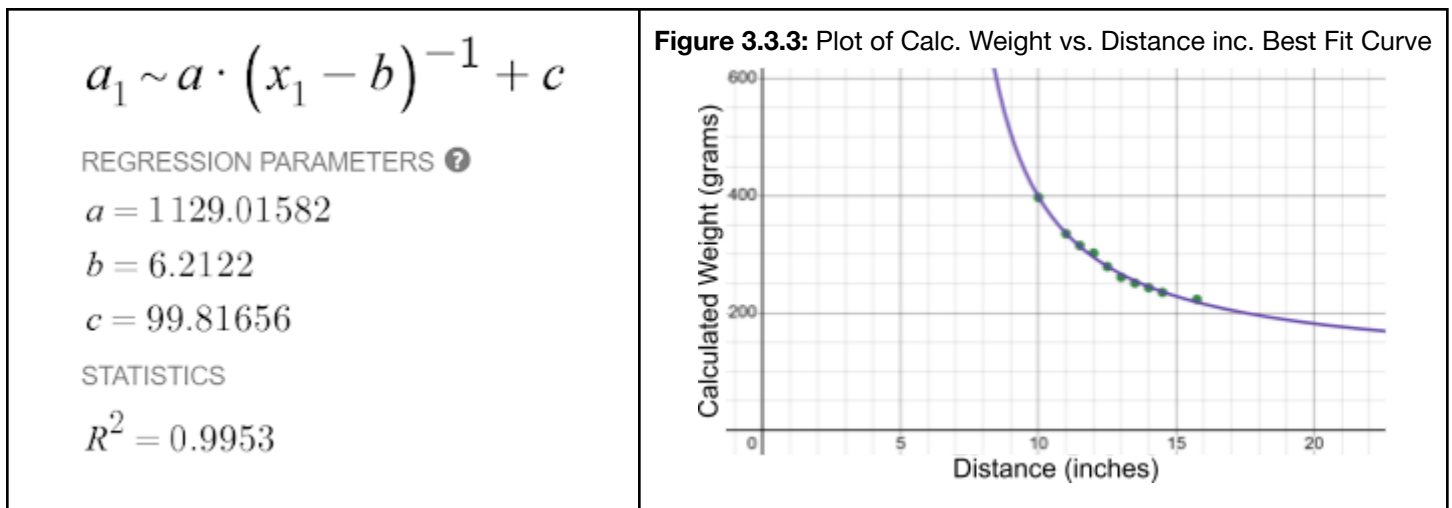
Figure 3.3.1: Image showing the testing setup for Test 3

Base information:

- Weight of glove / with controller = 362 g / 492 g
- Weight of glove with controller with box = 640 g
- Box length = 17 inches

Distance from edge of box (in)	Weight of glove with box and controller (g)	Weight of Box (g)	Calculated Weight Reading (g)
15.75	295	72	223
14.50	310	75	235
14.00	321	78	243
13.50	345	87	251
13.00	353	92	261
12.50	375	96	279
12.00	401	99	302
11.50	416	101	315
11.00	440	105	335
10.00	515	118	397

Figure 3.3.2: Table with Raw Test Data



$$\text{Weight} = 1129.01582 * (\text{Distance} - 6.2122)^{-1} + 99.81656$$

$$\text{Distance} = 1129.01582 / (\text{Weight} - 99.81656) + 6.2122$$

$$\text{Center of Gravity} = \text{Distance} / 2 \quad @ \quad \text{Weight} = \text{Mass} / 2$$

$$\text{Distance} = 1129.01582 / (246 - 99.81656) + 6.2122$$

$$\text{Distance} = 13.9354812417 \text{ in}$$

$$\text{Center of Gravity} = 6.96774062085 \text{ in from base}$$

Data Analysis:

Since the glove was difficult to remain straight and constant, a rigid box was used to help determine the center of gravity. A scale was then used to record the weight of the box given different distances from a pivot. As this distance increased the value of the weight increased. By using inverse regression, a function can be derived which accurately models the relationship between the data above. Using this function, a distance can be determined where the scale would report half the weight of the glove. This distance will be equal to exactly twice the distance from the edge of the glove to the center of gravity. This value was calculated to be 6.9677 inches. This value means that the center of gravity is located approximately 7 inches away from the end of the glove, will be located on the forearm, about 1 - 2 inches from the wrist. This value is acceptable because one of the goals of this project was to move the center of gravity from the center of the hand to the forearm, which has been accomplished, and accomplished well.

IV. Conclusion

A solution to a technical problem is inherently an iterative process. Regardless of how much time is spent coming up with an idea for a solution, there will always be unexpected problems that arise even when just physically making the idea. If the solution isn't being constantly tested and adjusted, the final result will fall short of the expected results.

The purpose of evaluating test results is to determine two things: the effectiveness of the product, and how to improve future products. Test results act as a quantifiable way to measure how well certain parts of a product perform. Designers can be confident that their product is effective if it manages to succeed in all of their tests. Shortcomings of a product can also be revealed if some of the tests fail. These failures and successes can then be used to determine where the designers should iterate the product so that it can be as successful as possible.